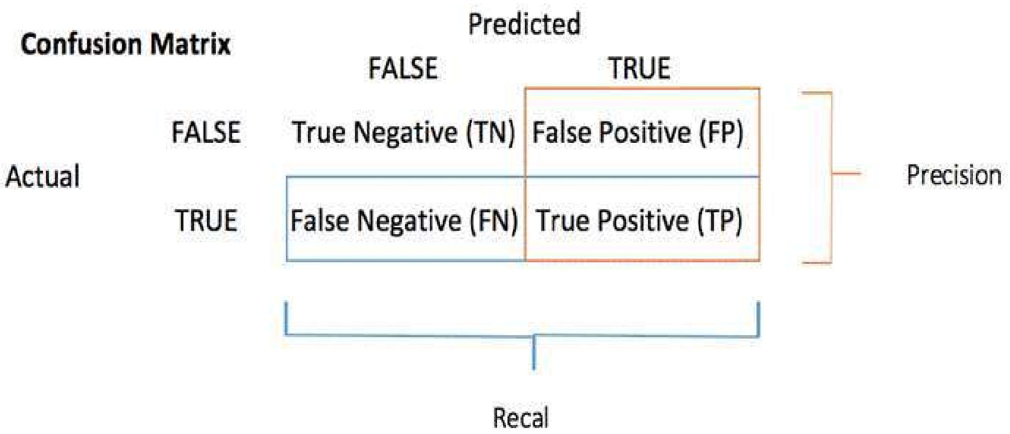
Confusion matrix:

table used in machine learning and statistics to assess the performance of a supervised classification model

A confusion matrix is a visual representation of the performance of a machine learning model.

confusion matrix, also known as an error matrix



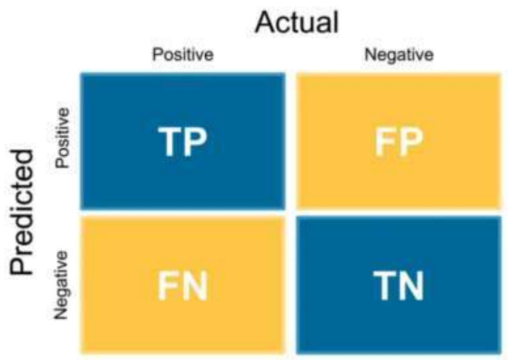
Each row of the matrix represents the instances in an actual class while each column represents the instances in a predicted class (or vice versa)

Confusion matrix signifies the predicted and actual values of a classification model to identify misclassifications.

The confusion matrix helps data scientists to fine-tune their models and improve their performance.

TECHNISCHE HOCHSCHULE DEGGENDORF
Campus Cham

Confusion matrix:



A confusion matrix is used with labeled datasets.

It's a table that compares the predicted labels of a classification model to the true labels (actual values) in the dataset.

This allows for an evaluation of the model's performance and the types of errors

True Positive (TP)
The model predicted true and it is true.
The model predicted that someone is sick and the person is sick.

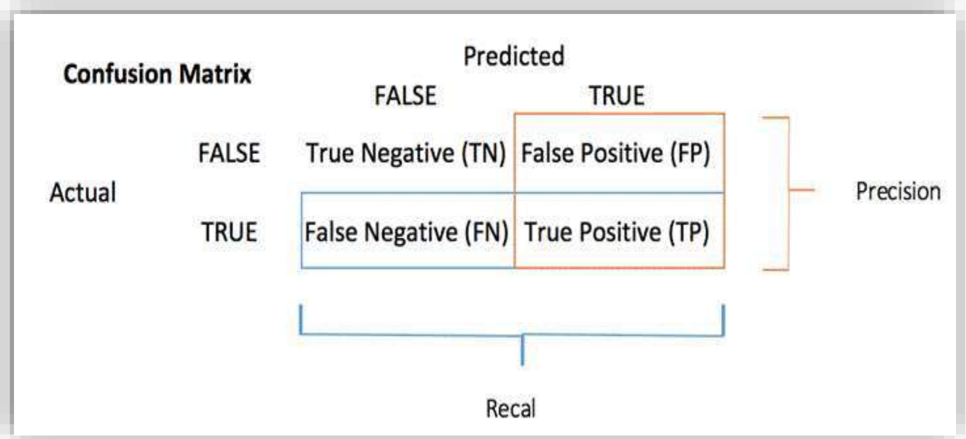
False Positive(FP)
The model predicted True and it is false.
The model predicted that someone is sick and the person is not sick.

True Negative
The model predicted false and it is false.
The model predicted that someone is not sick and the person is not sick.

False Negative
The model predicted false and it is true.
The model predicted that someone is not sick and the person is sick.

JC CHOUINARD

Confusion Matrix: Accuracy, Precision, Recall, F1-score



TN = True negative: Predicted values correctly predicted as an actual negative.
TP = True positive: Predicted values correctly predicted as actual positive.
FN = False negative: Positive values predicted as negative.
FP = False positive: Predicted values correctly predicted as an actual negative.

Key points:

- The diagonal elements (TP and TN) represent the correct predictions.
- Off-diagonal elements (FP and FN) represent incorrect predictions.
- A higher diagonal value indicates a better model performance.

Accuracy is a measure for the closeness of the measurements to a specific value.

Accuracy = (TN + TP) / (TN + TP + FN + FP)

Precision is the closeness of the measurements to each other, i.e. not necessarily to a specific value.

Precision = TP / (TP + FP)

Recall, also known as sensitivity, is the ratio of the correctly identified positive cases to all the actual positive cases, which is the sum of the "False Negatives" and "True Positives".

Recall = TP / (TP + FN)

F1 score is a weighted average score of the true positive (recall) and precision

F1-score = 2 (Precision x Recall) / (Precision + Recall)

Accuracy

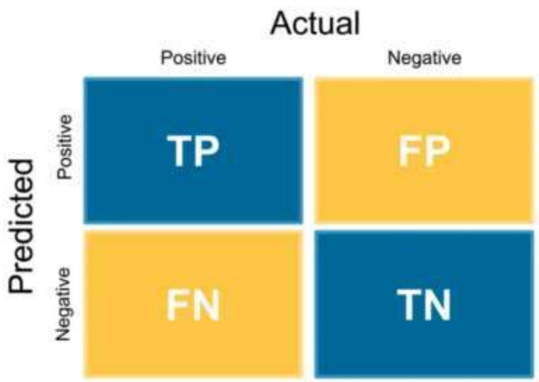
For classification

✓ Accuracy/Precision/Recall/F1-score

Accuracy is a measure of how the prediction from our model are close to their true value.

$$\text{Accuracy} = \frac{\#\ \text{Correct classifications}}{\#\ \text{All classifications}}$$

Example:
If a classifier make 10 predictions and 9 of them are correct, the accuracy is 90%.

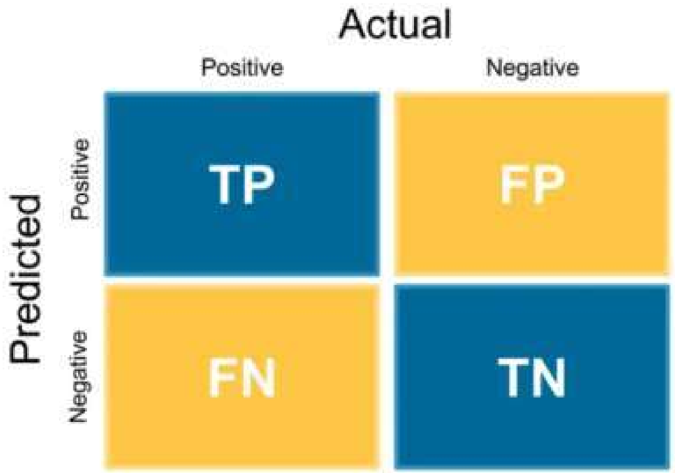


Accuracy = (TN + TP) / (TN + TP + FN + FP)

Precision

Precision (How much we are precise in the detection)

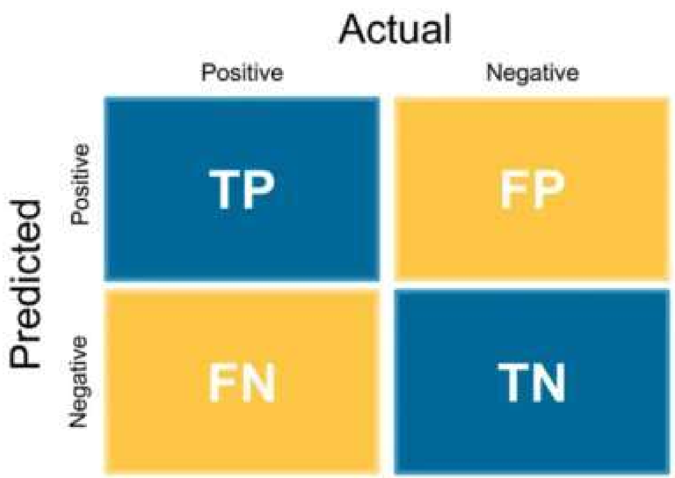
Precision is the closeness of the measurements to each other, i.e. not necessarily to a specific value.



Precision = TP / (TP + FP)

Recall

Recall (How much we are good at detecting)

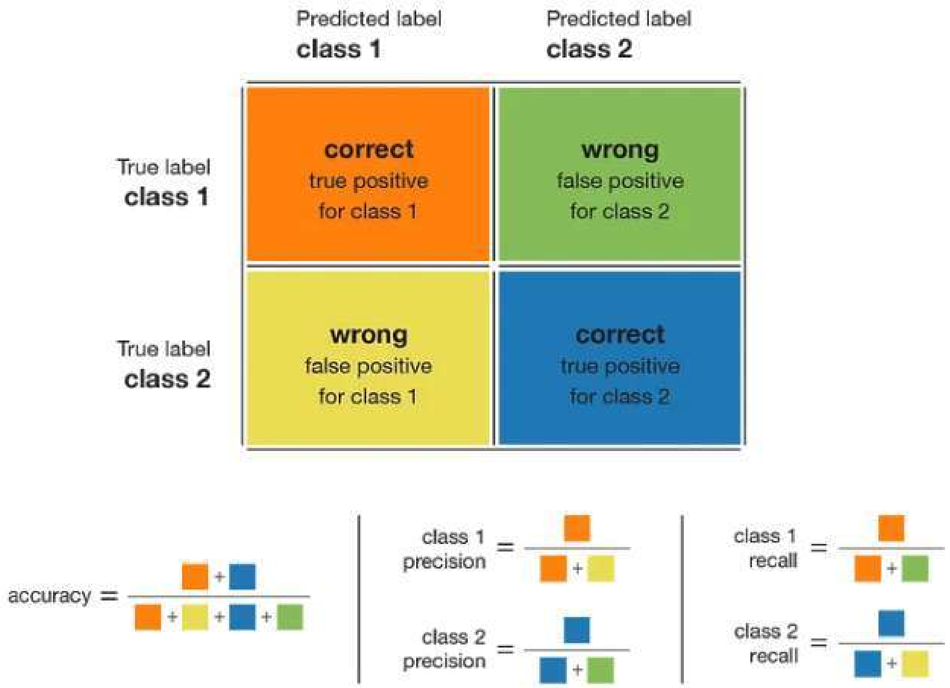


Of all patients that actually have the disease, what fraction did we correctly detect as having the disease?

Recall, also known as sensitivity, is the ratio of the correctly identified positive cases to all the actual positive cases, which is the sum of the "False Negatives" and "True Positives".

$$\frac{\text{True Positive}}{\#\ \text{Actual Positive}} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

Impact of Precision and Recall on Class



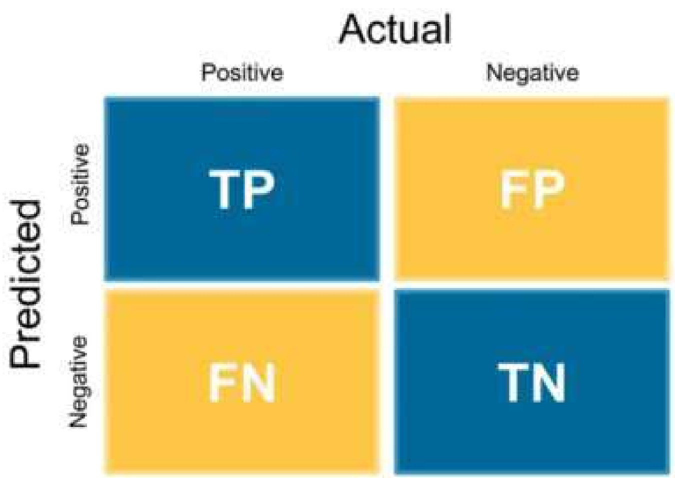
For a given class, the different combinations of recall and precision have the following meanings :

- high recall + high precision :
  The class is perfectly handled by the model
- low recall + high precision :
  The model can't detect the class well but is highly trustable when it does
- high recall + low precision :
  The class is well detected but the model also include points of other classes in it
- low recall + low precision :
  The class is poorly handled by the model

F1-score:

F1 score is a weighted average score of the true positive (recall) and precision

a general rule of thumb, an F1 score of 0.7 or higher is often considered good.



F1-score = 2 (Precision x Recall) / (Precision + Recall)

F1 score shows a strong performance in recognizing positive cases while minimizing false positives and false negatives.

This makes it a suitable metric when recall and precision must be optimized simultaneously, especially in imbalanced datasets.

Confusion matrix: Salient features

- ✓ It shows how any classification model makes predictions.
- ✓ Confusion matrix not only gives you insight into the errors being made by your classifier but also types of errors that are being made.
- ✓ This breakdown helps you to overcomes the limitation of using classification accuracy alone.

Confusion Matrix for 1000 products

	Predicted label **defective**	Predicted label **not defective**
True label **defective**	0	380
True label **not defective**	0	9620

$$\text{accuracy} = \frac{9620 + 0}{9620 + 380 + 0 + 0} \qquad \text{defective recall} = \frac{0}{0 + 380}$$

$$\text{not defective precision} = \frac{9620}{380 + 9620} \qquad \text{not defective recall} = \frac{9620}{9620 + 0}$$